

Machine Learning

机器学习

21/02/2025

Last time

- Correlation : pearson, kendal, ...
- Linear regression & logistic regression with statsmodel
- Evaluating the model
 - R-squared, p-values, coefficients, ...
 - ~~○ conditions for linear regression~~
- classification metrics
- bootstrap

Today

- Warmup
- Classification
 - [Imbalanced datasets](#)
 - [categorical variables](#)
 - multiclass classification
- from stats to Machine Learning (statsmodel to sklearn)
- [Machine Learning](#)
- [stochastic gradient descent](#)

Confusion matrix - (the name is fitting)

		True condition				
Total population		Condition positive	Condition negative	Prevalence = $\frac{\Sigma \text{ Condition positive}}{\Sigma \text{ Total population}}$	Accuracy (ACC) = $\frac{\Sigma \text{ True positive} + \Sigma \text{ True negative}}{\Sigma \text{ Total population}}$	
Predicted condition	Predicted condition positive	True positive, Power	False positive, Type I error	Positive predictive value (PPV), Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Predicted condition positive}}$	False discovery rate (FDR) = $\frac{\Sigma \text{ False positive}}{\Sigma \text{ Predicted condition positive}}$	
	Predicted condition negative	False negative, Type II error	True negative	False omission rate (FOR) = $\frac{\Sigma \text{ False negative}}{\Sigma \text{ Predicted condition negative}}$	Negative predictive value (NPV) = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Predicted condition negative}}$	
		True positive rate (TPR), Recall, Sensitivity, probability of detection $= \frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	False positive rate (FPR), Fall-out, probability of false alarm $= \frac{\Sigma \text{ False positive}}{\Sigma \text{ Condition negative}}$	Positive likelihood ratio (LR+) $= \frac{\text{TPR}}{\text{FPR}}$	Diagnostic odds ratio (DOR) $= \frac{\text{LR+}}{\text{LR-}}$	F ₁ score = $\frac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$
		False negative rate (FNR), Miss rate $= \frac{\Sigma \text{ False negative}}{\Sigma \text{ Condition positive}}$	Specificity (SPC), Selectivity, True negative rate (TNR) $= \frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Negative likelihood ratio (LR-) $= \frac{\text{FNR}}{\text{TNR}}$		

from Wikipedia [Confusion Matrix](#) page

What about the threshold ?

What happens if we use a different classification threshold ?

The confusion matrix and associated metrics also change.

t = 0.5	Predicted Default	Predicted Non-Default
Actual Default	286	47
Actual Non-Default	40	460

Acc (0.5) = 746 / 833 = 89.56%

TPR = 286 / 333 = 85.89%

FPR = 47 / 333 = 14.11%

t = 0.75	Predicted Default	Predicted Non-Default
Actual Default	243	90
Actual Non-Default	19	481

Acc (0.75)=(243 + 481)/888 = 81 %

TPR = 243 / 333 = 72.97%

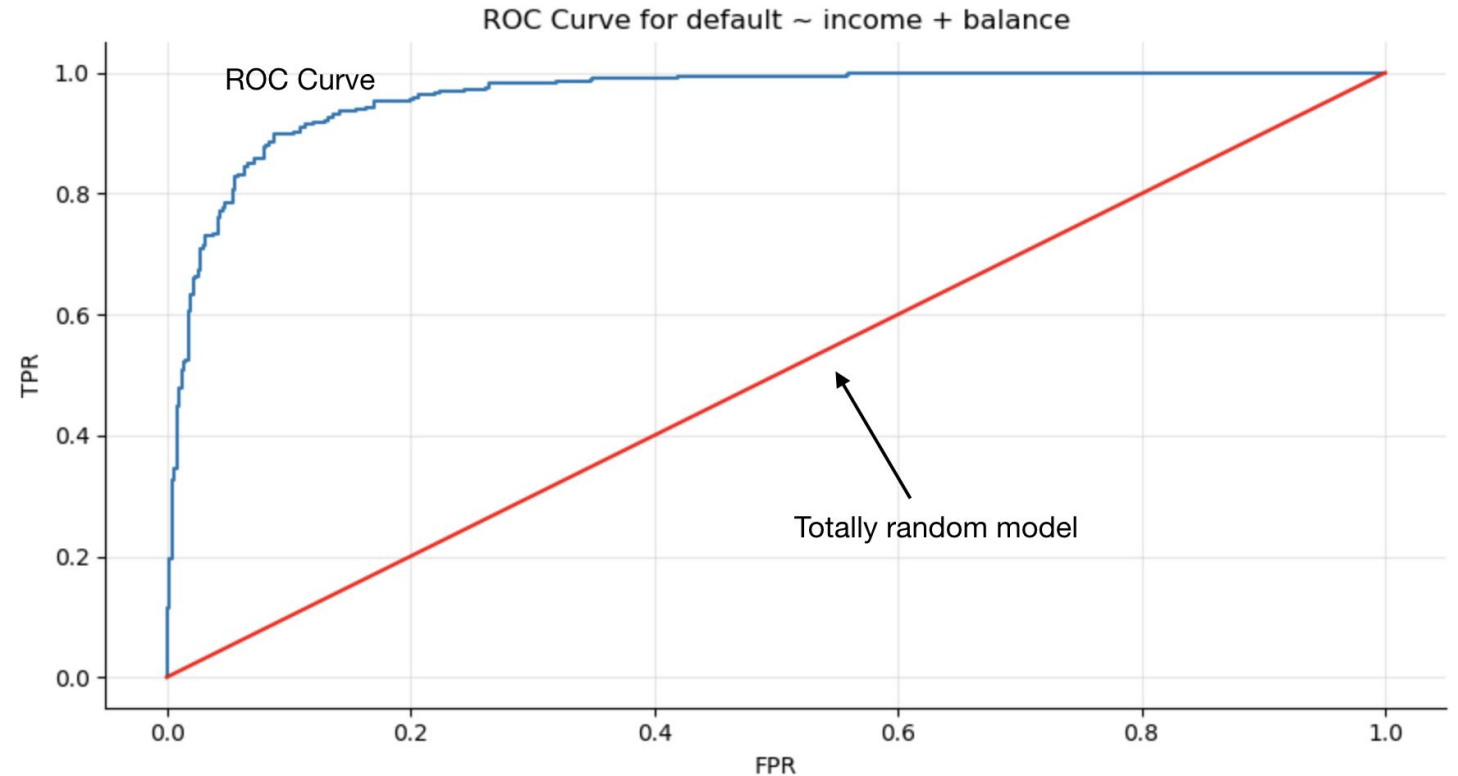
FPR = 90 / 333 = 27.03%

ROC-AUC

Plotting the TPR vs. FPR curve while varying the threshold from 0 to 1 gives the **ROC** curve.

The area under the curve (**AUC**) is a more robust metric than accuracy.

See [explanation](#) and [history](#)



```
yhat= results.predict(df)

from sklearn.metrics import roc_curve
fpr, tpr, thresholds = roc_curve(df['default'], yhat)

from sklearn.metrics import roc_auc_score
score = roc_auc_score(df['default'], yhat)
```

Warmup

Warmup (30mn)

Setup

In Google Colab

- Add your Colab to your name in the spreadsheet
- Add me as editor with my email alexis.perrier@gmail.com

Data

- Download the **credit_default** dataset from [github](#) and add it to your colab.

Build

Using statsmodel build the logistic regression model

- **default ~ income + balance + student**

- Interpret the results
- Compute the confusion matrix
- Calculate accuracy, TPR, and FPR
- Plot the histogram of probabilities for the prediction
- Compute AUC
- Plot the ROC curve

if you have time, compare with the simpler model **default ~ income**

Next

Imbalanced dataset & Categorical Encoding